One-Sided Extendability in the Four-Letter Abelian-Square-Free Language

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Abstract

Let $\mathcal{A}_4$ be the language of finite Abelian-square-free words over a four-letter alphabet. We exhibit the explicit word

$$s = \texttt{abacabadc}$$

which cannot be extended by even one letter to the left while remaining Abelian-square-free, yet is a prefix of a right-infinite Abelian-square-free word. The negative half is a four-case hand check. For the positive half we iterate the prefix-morphism operator $F(V) = Cg(V)$, where $C = \texttt{abacabadcdb}$ and g is Keränen's 85-uniform Abelian-square-free endomorphism. Boundary-crossing obstructions reduce to a finite exact certificate: 99 seed geometries, 35 residual states, and 17 recursive transition rows. All nonrecursive cases lie in a prefix of length 178 and are checked inside a conservative 190-symbol window. A separate clean-room implementation independently reconstructs the finite certificate. The witness realizes, in a particularly strong form, the one-sided-extension phenomenon posed by Keränen for unfavourable factors.

1 Introduction

An *Abelian square* is a word uv with $|u| = |v| > 0$ in which u and v have the same Parikh vector. Erdős asked whether an infinite word over four letters can avoid Abelian squares. Keränen answered this affirmatively by constructing an 85-uniform morphism g which is itself Abelian-square-free: if w is Abelian-square-free, then so is $g(w)$ [4].

The existence of infinite Abelian-square-free words does not imply that every finite Abelian-square-free word has an infinite continuation. Keränen later studied *unfavourable factors*, finite Abelian-square-free words which cannot occur as proper factors of an infinite Abelian-square-free word. In the 2010 account he noted that such a factor might nevertheless be extendable without bound in one direction, and stated that the existence of such factors remained open at that time [5].

We prove the following concrete asymmetric phenomenon.

Main theorem (informal form). The word

$$s = \texttt{abacabadc}$$

is a prefix of a right-infinite Abelian-square-free word, but no letter can be prepended to s without creating an Abelian square.

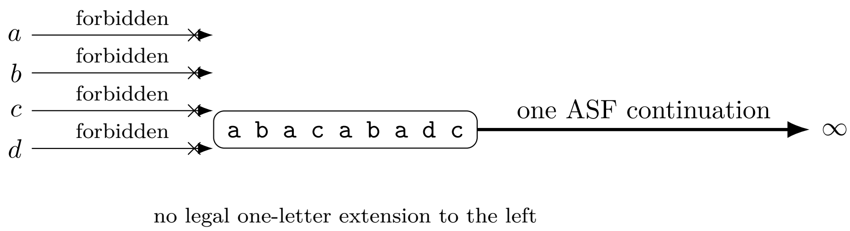

Figure 1: The theorem in one picture. Every possible first step to the left is illegal, while at least one Abelian-square-free continuation exists indefinitely to the right. The left obstruction is verified explicitly in Section 3; the right arrow is only a schematic summary of the construction proved in Sections 4–8.

What the theorem says. The left side of s is completely blocked: there is no legal first letter to add. The right side behaves in the opposite way: one can continue forever while avoiding every Abelian square. This is an extreme one-sided asymmetry. The left-hand obstruction is local and elementary; the right-infinite continuation is global and is proved by combining Keränen's morphism with a finite exact boundary certificate.

No claim of minimality is made for the length-9 witness. Correctness is independent of historical priority: Keränen's 2010 paper explicitly posed the one-sided-extension phenomenon as open at that time, while any claim about later priority is bibliographic rather than mathematical.

2 Preliminaries and notation

Let $\Sigma_4 = \{a, b, c, d\}$. For a finite word u, write

$$P(u) = (|u|_a, |u|_b, |u|_c, |u|_d)$$

for its Parikh vector. Words u, v are *Abelian equivalent* if $P(u) = P(v)$. An *Abelian square* is a factor uv with $|u| = |v| > 0$ and $P(u) = P(v)$. Let $\mathcal{A}_4$ denote the factorial language of finite words over Σ_4 containing no Abelian square.

Following Shur [7], the right-, left-, and two-sided extendable parts of a language L consist of words having arbitrarily long corresponding extensions in L:

$$\begin{aligned} w \in re(L) &\iff \forall n\ \exists v,\ |v| \geq n,\ wv \in L, \\ w \in le(L) &\iff \forall n\ \exists u,\ |u| \geq n,\ uw \in L, \\ w \in e(L) &\iff \forall n\ \exists u, v,\ |u|, |v| \geq n,\ uwv \in L. \end{aligned}$$

For a factorial language,

$$e(L) \subseteq re(L) \cap le(L).$$

When a nested infinite right extension is explicitly constructed, membership in $re(L)$ is immediate. Conversely, for a factorial language over a finite alphabet, arbitrarily long compatible finite right extensions yield a one-sided infinite extension by the usual finitely branching tree argument: factoriality makes the continuation tree prefix-closed, finite alphabet makes it finitely branching, and König's lemma supplies an infinite branch.

The informal theorem from Section 1 can therefore be written as

$$\boxed{s \in re(\mathcal{A}_4) \setminus le(\mathcal{A}_4)},$$

and consequently

$$\boxed{re(\mathcal{A}_4) \setminus e(\mathcal{A}_4) \neq \varnothing}.$$

Symbol	**Meaning**
Σ_4	alphabet $\{a, b, c, d\}$
$P(w)$	Parikh vector of w
$\mathcal{A}_4$	finite Abelian-square-free language on Σ_4
s	witness `abacabadc`
C	boundary word `abacabadcdb`
g	Keränen's 85-uniform Abelian-square-free morphism
$F(V)$	prefix-morphism operator $Cg(V)$
$W_n,\ W_\infty$	finite nested approximants and their right-infinite limit
$R(q, x, y)$	residual occurrence recording a Parikh imbalance and two marked letters
Q	the finite set of 35 residual states
$re,\ le,\ e$	right-, left-, and two-sided extendable parts

Table 1: Notation used throughout the proof.

3 A word which is dead on the left

Set

$$s = \texttt{abacabadc}.$$

A direct check shows that $s \in \mathcal{A}_4$.

Before the four cases, note that an Abelian square need not repeat the same word literally. For example,

$$\texttt{dabac} \mid \texttt{abadc}$$

has different halves, but each half contains two a's and one each of b, c, d. Their Parikh vectors are equal, so the factor is an Abelian square.

Lemma 3.1. *No letter in Σ_4 can be prepended to s while preserving Abelian-square-freeness.*

Proof. The following Abelian squares occur in the four possible extensions.

prepended letter	half-period	forbidden factor	Parikh vector
a	1	$a \mid a$	$(1, 0, 0, 0)$
b	2	$ba \mid ba$	$(1, 1, 0, 0)$
c	4	$caba \mid caba$	$(2, 1, 1, 0)$
d	5	$dabac \mid abadc$	$(2, 1, 1, 1)$

Thus no length-1 left extension belongs to $\mathcal{A}_4$. Any nonempty left extension us would have a suffix xs, where $x \in \Sigma_4$ is the last letter of u; factoriality would then force $xs \in \mathcal{A}_4$, a contradiction. Hence $s \notin le(\mathcal{A}_4)$. $\square$

4 Keränen's morphism and the right-infinite construction

We use Keränen's 85-uniform Abelian-square-free endomorphism g [4]. Its four images are given in Appendix A. The classical property used here is

$$w \in \mathcal{A}_4 \quad \Longrightarrow \quad g(w) \in \mathcal{A}_4. \tag{4.1}$$

Let

$$C = \texttt{abacabadcdb}$$

and define $F(V) = C\,g(V)$. Starting from $W_0 = C$, put

$$W_{n+1} = F(W_n) = C\,g(W_n).$$

Since g is a morphism,

$$W_n = C\,g(C)\,g^2(C)\cdots g^n(C), \tag{4.2}$$

so W_n is a prefix of W_{n+1}. Hence the limit

$$W_\infty = C\,g(C)\,g^2(C)\cdots \tag{4.3}$$

is well-defined. The only possible new squares under F are those meeting the prefix C, because factors wholly contained in $g(V)$ are covered by (4.1).

Roadmap for the right-infinite proof

The remaining proof has four ingredients.

1. Keränen's theorem makes the interior of $g(V)$ safe whenever V is Abelian-square-free.
2. A large square crossing the new boundary $C \mid g(V)$ forces one of finitely many source-level residual states.
3. A residual state occurring far enough to the right in $Cg(V)$ desubstitutes to another residual state in V.
4. All short or nonrecursive configurations lie in one fixed prefix and are checked exactly.

These ingredients define an invariant class $\mathcal{C}^*$ with

$$C \in \mathcal{C}^*, \qquad V \in \mathcal{C}^* \Longrightarrow Cg(V) \in \mathcal{C}^*.$$

The nested sequence W_n then remains Abelian-square-free at every level.

5 Residual states: compressing a boundary obstruction

A boundary-crossing Abelian square compares two long halves. Most of each half consists of complete g-images. Complete images contribute only through their Parikh vectors, so after cancellation the source-level information that matters is much smaller than the square itself: a Parikh imbalance between a source prefix and an intervening gap, together with the two source letters whose image blocks contain the relevant cuts. A *residual state* records exactly this reduced information.

We use row Parikh vectors. Let M be the incidence matrix of g: the row indexed by x is $P(g(x))$. Directly from the four images,

$$M = \begin{pmatrix} 19 & 21 & 27 & 18 \\ 18 & 19 & 21 & 27 \\ 27 & 18 & 19 & 21 \\ 21 & 27 & 18 & 19 \end{pmatrix}, \qquad \det M = 43435 \neq 0. \tag{5.1}$$

Thus $qM = r$ has at most one rational solution q, and integrality can be checked exactly.

Definition 5.1 (Residual occurrence)**.** For $q \in \mathbb{Z}^4$ and $x, y \in \Sigma_4$, a residual state $R(q, x, y)$ occurs in a word V if

$$V = A\,x\,B\,y\,D \tag{5.2}$$

for some words A, B, D, with the displayed occurrences of x and y distinct and in this order, and

$$P(A) - P(B) = q. \tag{5.3}$$

5.1 States forced by a large crossing square

Suppose an Abelian square in $F(V) = Cg(V)$ is not contained wholly in $g(V)$. Its start is at a position $i \in \{0, \dots, 10\}$ of C. Let its half-period be $K \geq 85$. We may write the relevant part of the source word as

$$V = A\,x\,B\,y\,D,$$

where the midpoint lies at offset $r \in \{0, \dots, 84\}$ of $g(x)$, and the end lies at prefix length $t \in \{0, \dots, 85\}$ of $g(y)$.

Equality of the two half-Parikh vectors gives

$$qM = P(g(x)[r:]) + P(g(y)[:t]) - P(C[i:]) - P(g(x)[:r]), \tag{5.4}$$

where $q = P(A) - P(B)$.

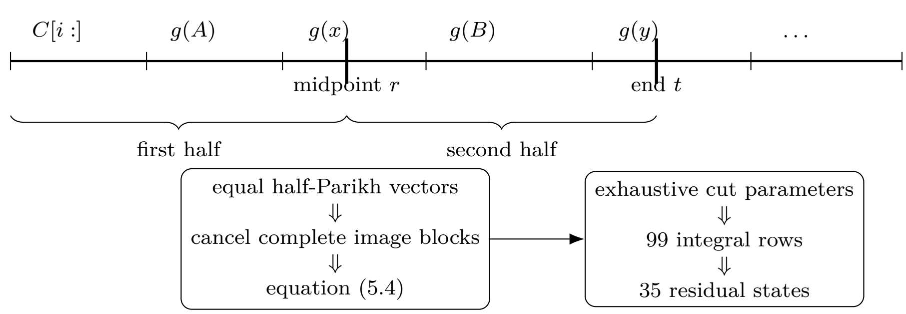

Figure 2: From a large boundary-crossing Abelian square to a residual state. The cut positions are exhaustive: $0 \leq i \leq 10$, $x, y \in \Sigma_4$, $0 \leq r \leq 84$, and $0 \leq t \leq 85$. Equation (5.4) has at most one rational q for each tuple because $\det M \neq 0$; retaining the integral solutions yields 99 parameter rows and 35 distinct states. The diagram explains the reduction, while exact enumeration supplies the finite counts.

The finite parameter ranges are therefore

$$0 \leq i \leq 10, \qquad x, y \in \Sigma_4, \qquad 0 \leq r \leq 84, \qquad 0 \leq t \leq 85.$$

For each tuple, (5.4) has at most one rational solution q because $\det M \neq 0$; we retain exactly the integral solutions. This produces 99 valid parameter rows and

$$|Q| = 35 \tag{5.5}$$

distinct residual states. Their complete list and the 99 geometric rows are supplied as machine-readable supplementary data. The endpoint cases $t = 0$ and $t = 85$ are included in the enumeration; neither contributes an additional integral row.

Lemma 5.2 (Large crossing-square reduction)**.** *If V is Abelian-square-free and $Cg(V)$ contains an Abelian square meeting C with half-period at least 85, then V contains a residual state from Q.*

Proof. Choose i, x, r, y, t from the actual square and set $q = P(A) - P(B)$. Equation (5.4) holds. Since $q \in \mathbb{Z}^4$, this actual configuration is among the exact integral cases used to define Q. Hence $R(q, x, y)$ occurs in V. □

6 Recursive closure: pulling a residual back through the morphism

If a residual state occurs sufficiently far to the right in $Cg(V)$, both marked target letters lie inside images of source letters. The occurrence can then be pulled back through the morphism. Equation (6.1) computes the source-level residual produced by this *desubstitution*.

Suppose $R(q, x, y) \in Q$ occurs in $Cg(V)$ and the two marked target letters lie in distinct image blocks $g(h)$ and $g(k)$, at offsets r and t. Write the corresponding source factorization as

$$V = A\,h\,B\,k\,D,$$

and set $q' = P(A) - P(B)$. Expanding the target prefix and gap gives

$$q'M = q - P(C) - P(g(h)[:r]) + P(g(h)[r+1:]) + P(g(k)[:t]). \tag{6.1}$$

The omission of the marked target letter itself explains the suffix $g(h)[r+1:]$.

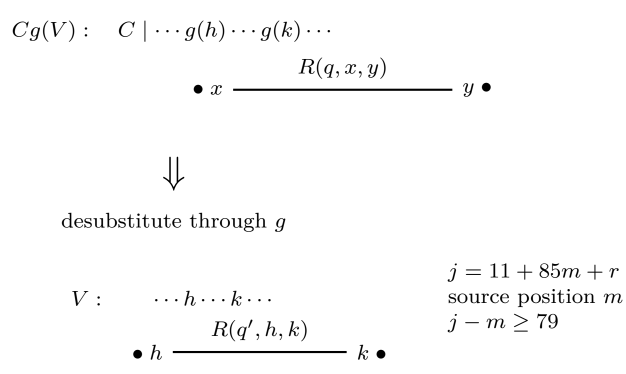

Figure 3: A recursive residual occurrence in $Cg(V)$ pulls back to a residual occurrence in V. The 17 exact integral transition rows cover all compatible target states, source letters, and image offsets. Although the state-label graph has a two-state cycle, every realized transition strictly decreases the first marked position; the minimum certified decrease is 79.

6.1 One concrete transition row

Consider the target state

$$R((0,1,0,-1),d,b)$$

with offsets $r = 68$ in $g(h)$ and $t = 63$ in $g(k)$, where $h = k = b$. The source factorization is $V = A\,b\,B\,b\,D$. Equation (6.1) uses

$$\begin{aligned} q &= (0,1,0,-1), & P(C) &= (4,3,2,2),\\ P(g(b)[:68]) &= (13,16,18,21), & P(g(b)[69:]) &= (5,3,3,5),\\ P(g(b)[:63]) &= (12,15,17,19). \end{aligned}$$

Consequently,

$$\begin{aligned} q'M &= (0,1,0,-1) - (4,3,2,2) - (13,16,18,21)\\ &\qquad + (5,3,3,5) + (12,15,17,19)\\ &= (0,0,0,0). \end{aligned}$$

Since $\det M = 43435 \neq 0$, this gives $q' = (0,0,0,0)$. Thus this transition row says, concretely, that the target occurrence desubstitutes to

$$R((0,0,0,0),b,b)$$

in V. The arithmetic displayed above is paper-and-pencil verifiable; the finite certificate records this row among all compatible alignments.

Put $m = |A|$. The first source mark is at position m, while the first target mark is at

$$j = |C| + 85m + r = 11 + 85m + 68 = 79 + 85m.$$

Hence

$$j - m = 79 + 84m \geq 79. \tag{6.2}$$

The same exact calculation is performed over every compatible target state, source letters, and image offsets. Exact inversion of (6.1) leaves exactly 17 integral parameter rows, and every resulting source triple (q',h,k) belongs again to Q.

Lemma 6.1 (Recursive residual closure)**.** *If a state from Q occurs in $Cg(V)$ with its marked letters in distinct image blocks beyond C, then a state from Q occurs in V.*

Proof. For an actual occurrence, the source vector $q' = P(A) - P(B)$ is integral and satisfies (6.1). The exact finite inversion therefore includes the actual alignment; every integral source state produced by that enumeration lies in Q. □

7 Base cases and strict descent

The recursive lemma applies only when the marked letters lie in distinct image blocks beyond C. Every remaining short configuration is bounded explicitly.

Lemma 7.1 (Fixed base window)**.** *Every crossing Abelian square of half-period $K < 85$, and every residual occurrence from Q which does not fall under the recursive residual lemma, is contained in the first 178 letters of $Cg(V)$, provided that V begins with C. The finite certificate checks the larger fixed prefix of length 190.*

Proof. There are three cases.

Short crossing square. A crossing square starts at a zero-based position $i \le 10$. If $K < 85$, its exclusive end is at most

$$10 + 2 \cdot 84 = 178.$$

First residual mark in C. For a residual occurrence

$$Cg(V) = A\,x\,B\,y\,D$$

with marked positions $j < k$, taking coordinate sums in $q = P(A) - P(B)$ gives

$$\sum q = j - (k - j - 1) = 2j + 1 - k,$$

so

$$k = 2j + 1 - \sum q. \tag{7.1}$$

The 35-state certificate satisfies

$$\sum q \in \{-1, 0, 1\} \tag{7.2}$$

for every q in Q. If $j \le 10$, then $k \le 22$, so these cases lie in a prefix of length 23.

Both residual marks in one image block. Put $d = k - j \le 84$. Equation (7.1) gives $j = d + \sum q - 1 \le 84$, and then $k \le 168$. These cases lie in a prefix of length 169.

Thus the three sufficient prefix lengths are

$$178, \qquad 23, \qquad 169,$$

whose maximum is

$$\boxed{178}.$$

The certificate deliberately checks the conservative 190-symbol prefix. No minimality claim is made for the 190-symbol window. □

The prefix condition used below is load-bearing. If V begins with C, then the first 190 letters of $Cg(V)$ are independent of the rest of V: they equal the first 190 letters of

$$W_1 = Cg(C).$$

The exact verifier checks that this fixed word is Abelian-square-free and contains no state from Q. Thus one finite scan genuinely covers every nonrecursive case, rather than merely one observed tower prefix.

The recursive step is also well-founded. If the first marked position in a recursive target occurrence is $j \ge 11$, its containing source letter occurs at position

$$a = \left\lfloor \frac{j - 11}{85} \right\rfloor$$

of V, hence $a < j$. Recursive desubstitution strictly decreases a nonnegative integer and cannot continue indefinitely. The clean-room reconstruction evaluates all 17 rows and finds a minimum exact margin of 79; it finds no nondecreasing realized transition.

If positions are forgotten, the state graph has one cyclic strongly connected component,

$$\{R((0,0,-1,0), c, b), R((0,1,0,-1), d, b)\}.$$

This state-only cycle does not threaten termination: a graph edge can be realized only together with its image-block position, and that position strictly decreases.

8 The invariant and the infinite word

Define

$$\mathcal{C}^* = \{V : C \text{ is a prefix of } V,\ V \in \mathcal{A}_4,\ \text{and } V \text{ contains no state from } Q\}. \tag{8.1}$$

The condition $C \preceq V$ is not cosmetic: it is exactly what makes the short prefix of every $Cg(V)$ universal and therefore checkable once and for all.

Proposition 8.1 (Invariance)**.** *If $V \in \mathcal{C}^*$, then $Cg(V) \in \mathcal{C}^*$.*

Proof. The word $Cg(V)$ begins with C.

Since V is Abelian-square-free, (4.1) implies that $g(V)$ is Abelian-square-free. Therefore any Abelian square in $Cg(V)$ must meet C. If its half-period is below 85, the fixed-base lemma places it in the checked 190-letter window. If its half-period is at least 85, the large crossing-square lemma would force a state from Q in V, contrary to $V \in \mathcal{C}^*$.

It remains to exclude states from Q in $Cg(V)$. A nonrecursive occurrence lies in the fixed base window and is excluded by the certificate. A recursive occurrence would, by recursive residual closure, force a state from Q in V, again a contradiction. Thus $Cg(V) \in \mathcal{C}^*$. ☐

What has now been proved. There is no unbounded case left untreated: squares inside $g(V)$ are excluded by Keränen's theorem; short boundary configurations lie in the fixed finite window; large crossing squares force a forbidden residual in V; and recursive residuals likewise pull back to a forbidden residual in V. This is the universal closure needed for every level of the construction.

Proposition 8.2. *The word C belongs to $\mathcal{C}^*$.*

Proof. The exact finite verifier checks directly that C is Abelian-square-free and contains none of the 35 states from Q. ☐

Corollary 8.3. *Every W_n is Abelian-square-free, and the limit W_∞ is a right-infinite Abelian-square-free word.*

Proof. The preceding propositions imply $W_n \in \mathcal{C}^* \subseteq \mathcal{A}_4$ for all n. Any finite factor of the nested limit W_∞ occurs in some W_n, so the limit contains no Abelian square. ☐

9 Main result

Theorem 9.1. *For $s = \texttt{abacabadc}$,*

$$\boxed{s \in re(\mathcal{A}_4) \setminus le(\mathcal{A}_4)}.$$

Proof. The word s is a prefix of C, hence a prefix of the right-infinite Abelian-square-free word W_∞. Thus $s \in re(\mathcal{A}_4)$. The left-death lemma gives $s \notin le(\mathcal{A}_4)$. ☐

Corollary 9.2.

$$re(\mathcal{A}_4) \setminus e(\mathcal{A}_4) \neq \varnothing.$$

Proof. Since $e(\mathcal{A}_4) \subseteq le(\mathcal{A}_4)$, the theorem gives the claim. ☐

10 Computer-assisted proof and reproducibility

The proof uses one classical external input, Keränen's theorem (4.1), and one finite exact computation specific to the boundary C. The author-side script `verify_p7_main_theorem_v4.py` recomputes from the displayed formulas:

1. the four immediate left-death witnesses;
2. the incidence matrix M and exact integer inversion;
3. the 99 large-square seed alignments;
4. the 35 unique residual states Q;
5. the 17 integral recursive transition rows and closure back into Q;
6. the bound $\sum q \in \{-1, 0, 1\}$;
7. absence of Abelian squares and residual states in the fixed 190-letter base window;
8. membership of C in the invariant.

It also performs direct regression checks on W_1 and on boundary-crossing squares in W_2. These finite long-word checks are not used as extrapolation in the all-depth proof.

A separate clean-room implementation, `P7_CODEX_CLEANROOM_VERIFIER.py`, was written without importing or parsing the author verifier or the submitted CSV certificates during reconstruction. Starting only from the displayed boundary word and morphism images, it independently reconstructs the same 99 seed geometry rows, 35 residual states, and 17 recursive transition rows, with exact set equality to the supplied data. It also reproduces the minimum descent margin 79, the unique two-state state-only cyclic component, the absence of any nondecreasing realized transition, and the independently sufficient base-prefix length 178. This is an independent computational reconstruction, not human peer review.

Obligation	**Source**
g_{85} preserves ASF	Keränen 1992
left death of s	direct finite proof
seed completeness	finite exact reconstruction
residual closure	finite exact reconstruction
short/base cases	direct finite check
infinite limit	mathematical induction

The exact state and transition tables are supplied as machine-readable supplementary files.

11 Relation to previous work

Keränen's 1992 construction established an Abelian-square-free endomorphism on four letters and therefore infinite four-letter Abelian-square-free words [4]. Carpi later showed, among other results, exponential growth in the number of four-letter Abelian-square-free words [2].

Finite maximality has a distinct literature. Cummings and Mays constructed finite maximal Abelian-square-free words [3]; Korn gave short maximal constructions and bounds [6], later improved by Bullock [1]. Bullock explicitly distinguishes left- and right-maximal words. These papers concern finite blocking; in this literature search we found no statement combining left maximality with an infinite Abelian-square-free continuation on the opposite side.

Keränen's later work on unfavourable factors is closer to the present question. In his 2010 account he observed that an unfavourable factor might nevertheless be extendable without bound in one direction and stated that the existence of such factors remained open at that time [5]. The main theorem realizes this phenomenon in a particularly strong form: the witness has a right-infinite continuation while admitting no legal one-letter extension on the left.

Shur's framework makes the set-theoretic consequence transparent: despite general results comparing growth rates of extendable parts of factorial languages, the actual sets $re(\mathcal{A}_4)$ and $e(\mathcal{A}_4)$ need not coincide [7].

To our knowledge, and based on primary-source, forward-citation, and alternate-term searches through 3 September 2026, no later result equivalent to the main theorem was located. This is a statement about the literature search, not a proof of priority; we therefore make no absolute priority claim.

12 Discussion

The witness is small and its obstruction is completely local on the left, whereas the proof of infinite survival on the right is global and computer-assisted. Natural further questions include the minimum possible witness length, structural classification of left-unextendable/right-infinite words, whether a shorter human proof of the right-infinite construction exists, and whether analogous one-sided infinite extendability occurs in related ternary almost-Abelian-square-free languages. No minimality or uniqueness claim is made for s, C, or the construction.

A The 85-uniform morphism

$g(a) =$ `abcacdcbcdcadcdbdabacabadbabcbdbcbacbcdcacbabdabacadcbcdcacdbcbacbcdcacdcbdcdadbdcb`
`ca`
$g(b) =$ `bcdbdadcdadbadacabcbdbcbacbcdcacdcbdcdadbdcbcabcbdbadcdadbdacdcbdcdadbdadcadabacadc`
`db`
$g(c) =$ `cdacabadabacbabdbcdcacdcbdcdadbdadcadabacadcdbcdcacbadabacabdadcadabacabadbabcbdbad`
`ac`
$g(d) =$ `dabdbcbabcbdcbcacdadbdadcadabacabadbabcbdbadacdadbdcbabcbdbcabadbabcbdbcbacbcdcacba`
`bd`

B Supplementary verification material

The supplementary proof data are `P7_V2_RESIDUAL_STATES.csv` (35 states), `P7_V2_SEED_ROWS.csv` (99 seed rows), `P7_V2_RECURSIVE_TRANSITIONS.csv` (17 recursive rows), `verify_p7_main_theorem_v4.py` (author-side finite verifier/reconstructor), `P7_CODEX_CLEANROOM_VERIFIER.py` (independent clean-room reconstruction), and `P7_CODEX_HIGH_CERTIFICATE_RECONSTRUCTION.md` (reconstruction equations, bounds, outputs, and digests).

The clean-room reconstruction does not import or parse the author verifier or submitted CSV certificate during reconstruction; it compares the independently reconstructed finite sets with the supplied data only after reconstruction. A SHA-256 manifest accompanies the release package.

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